

Homework (Jalapeno)

Q1. Solve for x : $2x + 5 = 11$

- (A) $x = 2$
- (B) $x = 3$
- (C) $x = 4$
- (D) $x = 5$
- (E) $x = 6$

Q2. The Great Pyramid of Giza's original height h can be found using the formula $h = 2w - 10$, where w is the width. If $w = 15$, what is h ?

- (A) $h = 20$
- (B) $h = 25$
- (C) $h = 30$
- (D) $h = 35$
- (E) $h = 40$

Q3. The ancient Mesopotamians used the formula $A = \frac{1}{2}bh$ to calculate the area of a triangle. Solve for b : $A = 12$, $h = 4$

- (A) $b = 4$
- (B) $b = 5$
- (C) $b = 6$
- (D) $b = 7$
- (E) $b = 8$

Q4. In ancient Egypt, the ratio of the height of a pillar to its base was given by $\frac{h}{b} = \frac{3}{2}$. Solve for h : $b = 8$

- (A) $h = 10$
- (B) $h = 12$
- (C) $h = 14$
- (D) $h = 16$
- (E) $h = 18$

Q5. The ancient Greeks used the formula $V = \frac{1}{3}\pi r^2 h$ to calcu-

late the volume of a cone. Solve for r : $V = 12$, $h = 6$

- (A) $r = 2$
- (B) $r = 3$
- (C) $r = 4$
- (D) $r = 5$
- (E) $r = 6$

Q6. The Roman architect Vitruvius used the formula $C = 2\pi r$ to calculate the circumference of a circle. Solve for r : $C = 12$

- (A) $r = 1$
- (B) $r = 2$
- (C) $r = 3$
- (D) $r = 4$
- (E) $r = 5$

Q7. The ancient Chinese used the formula $A = \sqrt{s(s-a)(s-b)(s-c)}$ to calculate the area of a triangle. Solve for s : $A = 12$, $a = 4$, $b = 5$, $c = 6$

- (A) $s = 7$
- (B) $s = 8$
- (C) $s = 9$
- (D) $s = 10$
- (E) $s = 11$

Q8. The ancient Babylonians used the formula $P = a + b + c$ to calculate the perimeter of a triangle. Solve for c : $P = 15$, $a = 5$, $b = 6$

- (A) $c = 2$
- (B) $c = 3$
- (C) $c = 4$
- (D) $c = 5$
- (E) $c = 6$

Open Ended Questions (FRQ)

Q9. The ancient Egyptians built a pyramid with a base area of 100 square meters and a height of 50 meters. Use the formula $V = \frac{1}{3}Bh$ to calculate the volume of the pyramid. Show all work.

Q10. A ancient Greek statue has a circular base with a radius of 3 meters. The height of the statue is 6 meters. Use the formula $V = \pi r^2 h$ to calculate the volume of the statue. Show all work.

Chef's Tips

- Provide clear instructions for each question
- Encourage students to show all work for free response questions
- Emphasize the importance of correct notation and formatting